LAB EXPERIMENT - 12

Department	AIDS
Course Code / Course Name	ITA1409 - ETHICAL HACKING
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Q12. Port Scanning Tools

AIM

To use port scanning tools in the Kali Linux operating system to identify open, closed and filtered ports on a target system, and to understand the basic functions of the Nmap, Zenmap, Masscan, ss, netstat and lsof tools.

COMMANDS TO EXECUTE

Open the terminal and execute the following commands:

1. **Nmap Service Version Scan**

   The -sV option scans the target and detects the service and its version running on each open port.

   Command: `nmap -sV 192.168.56.1`

OUTPUT:



2. **Nmap TCP SYN (Stealth) Scan**

   The -sS option sends SYN packets to the target ports without completing the TCP connection. It requires root privileges, so sudo is used.

   Command: `sudo nmap -sS 192.168.56.1`

OUTPUT:



3. **Nmap Specific Port Scan**

   The -p option scans only the ports given in the command, here ports 21, 22, 80 and 443.

   Command: `nmap -p 21,22,80,443 192.168.56.1`

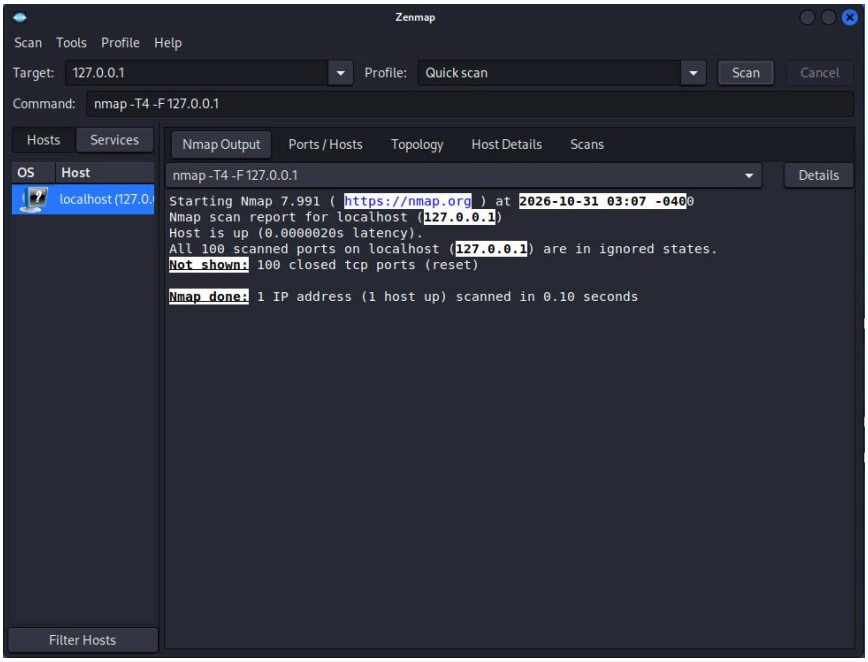
OUTPUT:



4. **Zenmap (Graphical Interface of Nmap)**

   Zenmap is the graphical front end of Nmap. A quick scan of the target 127.0.0.1 was performed using the command nmap -T4 -F 127.0.0.1.

OUTPUT:



5. **Masscan**

   Masscan is a very fast port scanner. The first command scans ports 22, 80 and 443 and the second scans ports 1 to 1000 of the target.

   Command: `sudo masscan 127.0.0.1 -p22,80,443`

   Command: `sudo masscan 127.0.0.1 -p1-1000`

OUTPUT:



6. **Listening Ports on the Local System (ss, netstat, lsof)**

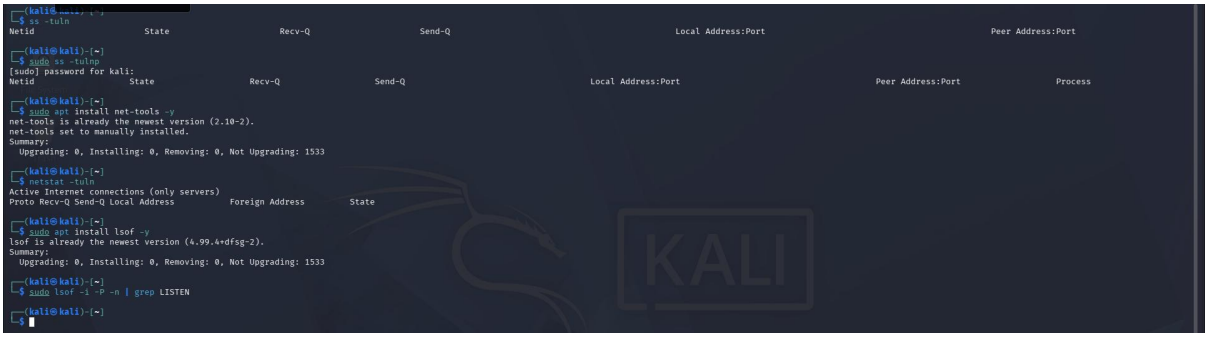
   The ss, netstat and lsof commands list the ports on which the local system is listening for connections. net-tools and lsof were already installed.

   Command: `sudo ss -tulnp`

   Command: `netstat -tuln`

   Command: `sudo lsof -i -P -n | grep LISTEN`

OUTPUT:



RESULT

The port scanning tools were successfully executed in Kali Linux. Nmap detected the open ports 135 (msrpc), 139 (netbios-ssn), 445 (microsoft-ds) and 3306 (mysql) on the target 192.168.56.1, with 996 closed ports, and reported ports 21, 22, 80 and 443 as filtered. Zenmap showed all 100 scanned ports of localhost as closed, Masscan discovered open ports 135 and 445 on 127.0.0.1, and the ss, netstat and lsof commands showed that no ports were listening on the Kali system.